

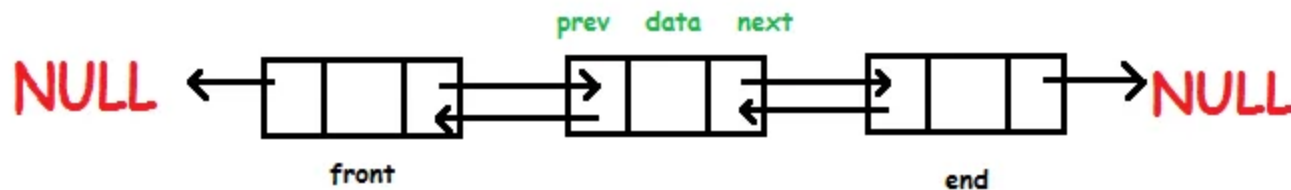


Doubly Linked List

1. Need of Doubly Linklist .
2. Doubly linked list
3. Circular Linked List.

Doubly linked list is a type of linked list in which each node apart from storing its data has two links. The first link points to the previous node in the list and the second link points to the next node in the list. The first node of the list has its previous link pointing to NULL similarly the last node of the list has its next node pointing to NULL.

This is also usefull for travelling linklist from both size



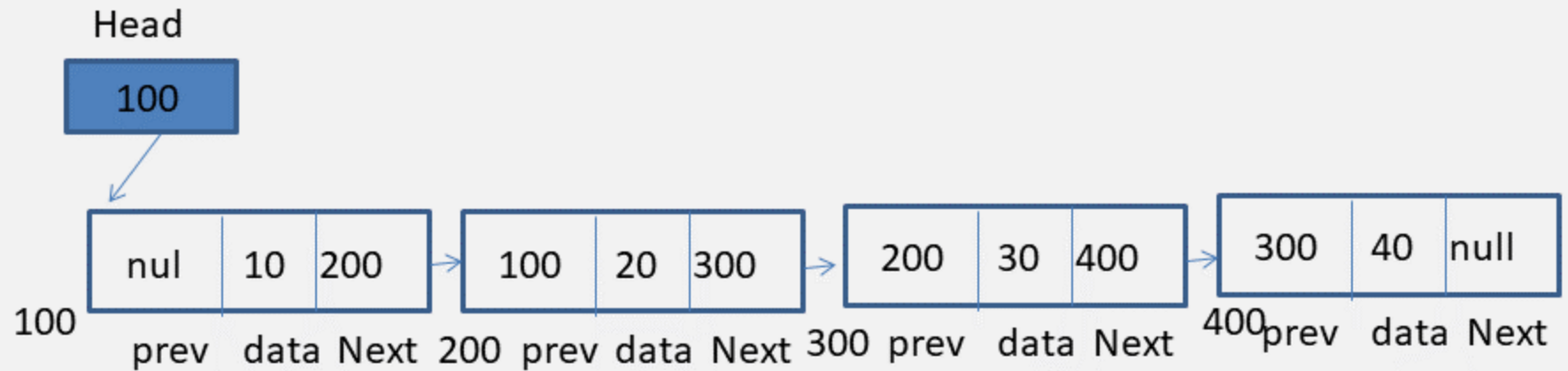
Implement doubly linked list with following menu.

1. Insert Node
2. Delete Node
3. Display Node
4. Modify Node
0. Exit

```
public class Node
{
    private int data;
    private Node prev,next;
    public Node()
    {
        data=0;
        prev=next=null;
    }
    public Node (int data )
    {
        this.data=data;
        prev=next=null;
    }
    public String toString()
    {
        return data;
    }
    // Getter and setter methods....
}
```

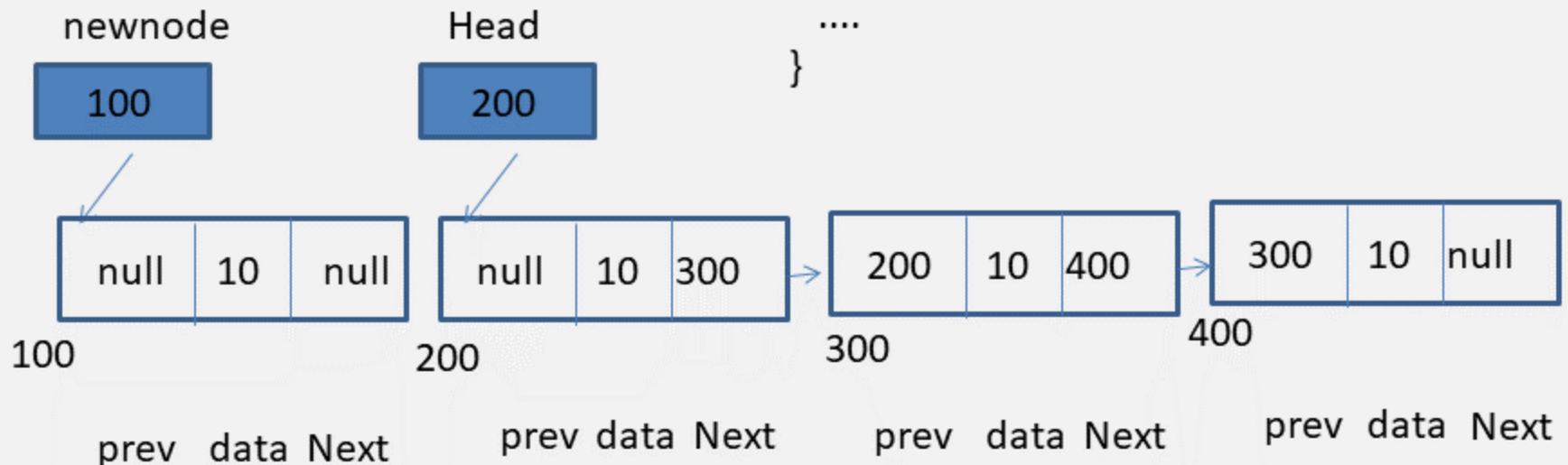
```
class linklist
{
    private Node head;
    public head()
    {
        head=null;
    }
    public void createLinklist(int data)
    { ... }
    public void insertNode(int data,int pos)
    { .... }
    public void modifyNode(int data)
    { .... }
    public void deleteNode(int pos)
    { ... }
    public String toString()
    { .... }
}
```

Doubly Linked List



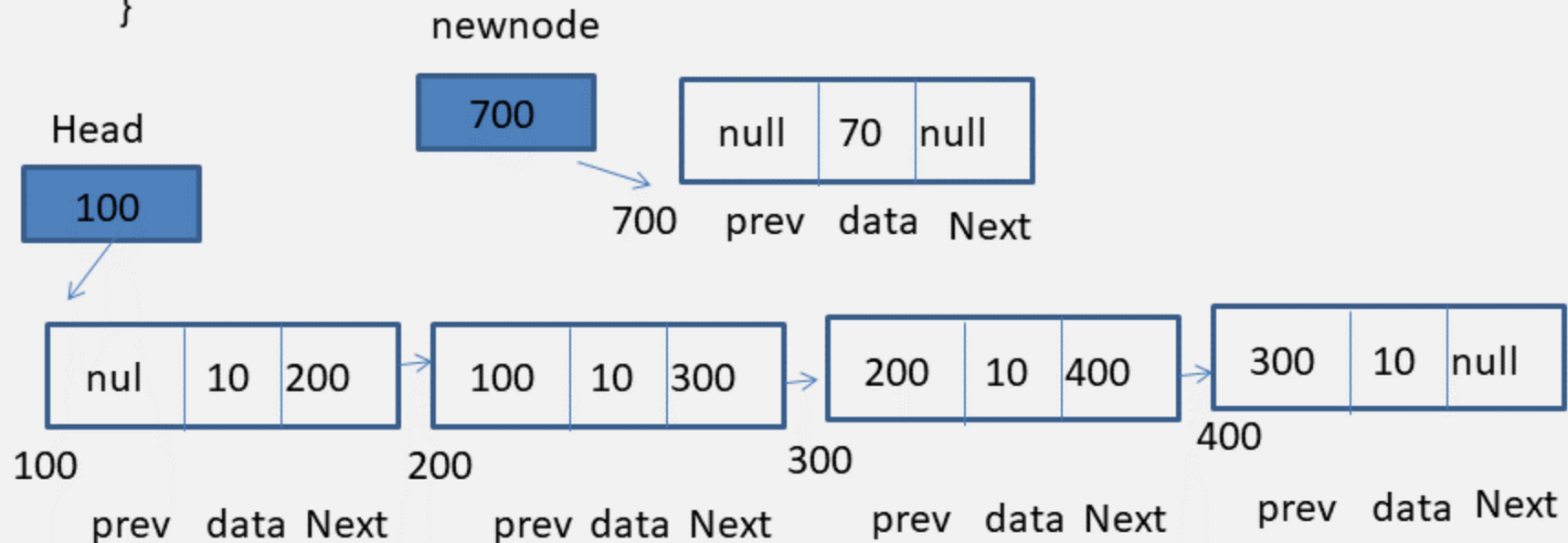
Insert Node ... Position-1

```
public void insert(int data,int pos)
{
    Node newnode = new Node(data);
    if ( pos==1)
    {
        newnode.setNext(head);
        head.setPrev(newnode);
        head=newnode;
    }
    ....
    ....
}
```



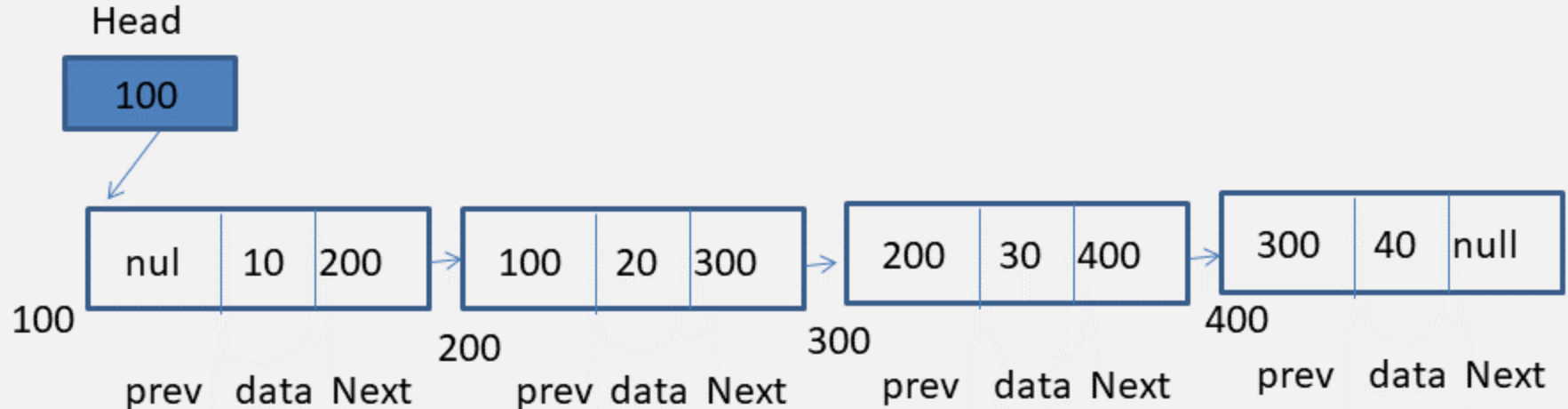
Insert Node ... Position other than 1

```
public void insertNode(int data,int pos) {  
    Node newnode=new Node(data);  
    if ( head ==null) {  
        head=newnode;  
        return ; }  
    if ( pos ==1)  
    {  
        head.setprev(newnode);  
        newnode.setNext(head);  
        head=newnode;  
    }  
    else {  
        Node move=head;  
        for ( int i=0; i < pos -1; i++)  
            move = move.getNext();  
        newnode.setNext(move.getNext());  
        if ( move.getNext() != null )  
            newnode.getNext().setPrev(newnode);  
        newnode.setPrev(move);  
        move.setNext(newnode);  
    }  
}
```

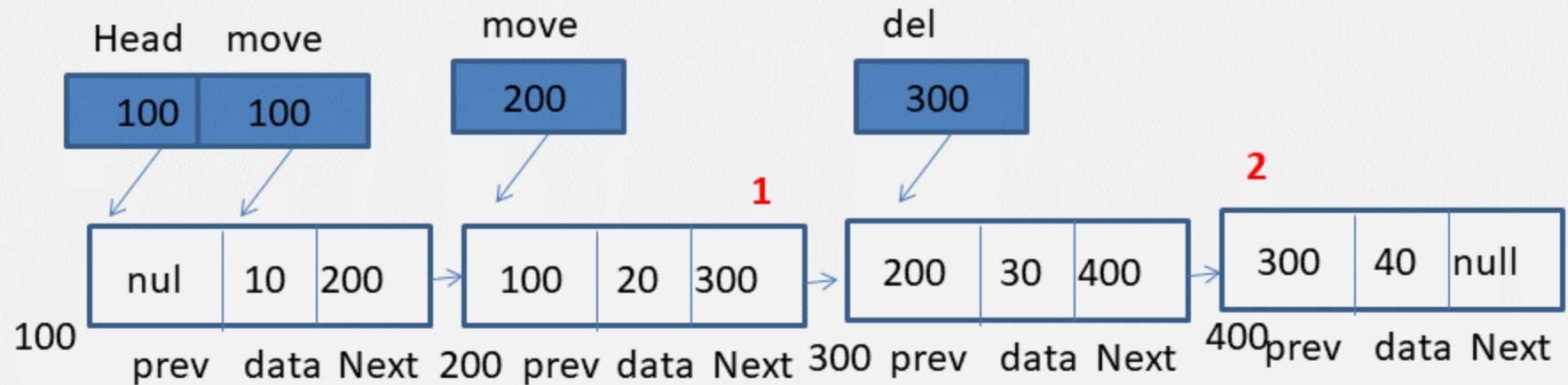


Delete Node (if position is 1)

- 1 Mark first node as del
del=head;
2. Move head to next node
head=head.getNext();
3. Assign prev as null for newly created head
head.setPrev(null).

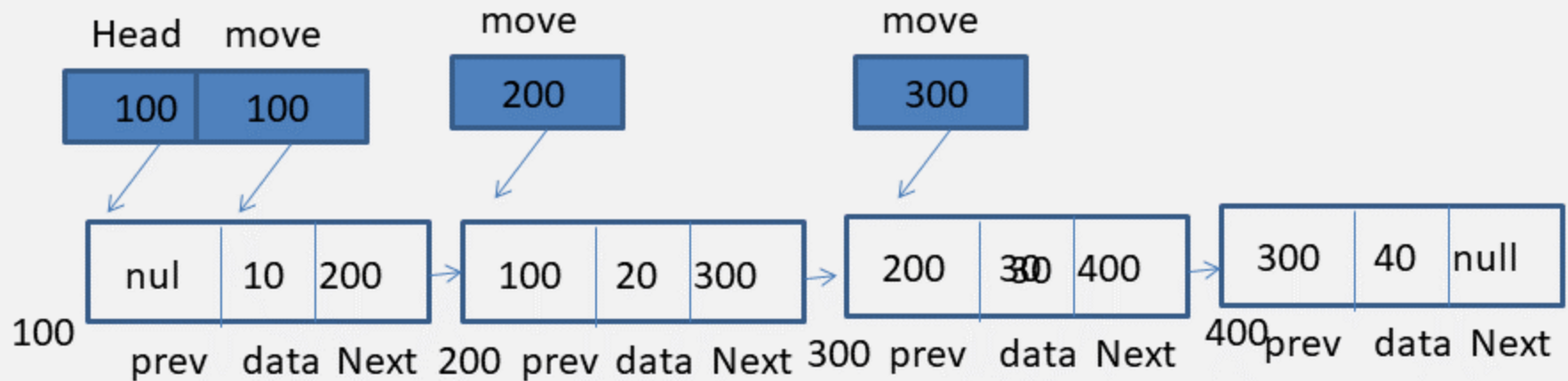


Delete Node (if position is other than 1)



Modify (Accept Data to Modify)

Data=30 new Data=35



```
if ( move.getdata()==data)
{
    move.setData(newdata);
    break;
}
move=move.getNext()
```

- ❖ Circular Linked List is a variation of Linked list in which the first element points to the last element and the last element points to the first element. Both Singly Linked List and Doubly Linked List can be made into a circular linked list.
- ❖ Circular linked list are mostly used in task maintenance in operating systems. There are many examples where circular linked list are being used in computer science including browser surfing where a record of pages visited in the past by the user, is maintained in the form of circular linked lists and can be accessed again on clicking the previous button.



